

Rukhsar Ahmed

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PROFESSIONAL SUMMARY

Senior Computer Vision Engineer with 5+ years of experience designing and deploying production-grade AI solutions. Specialized in deep Learning and Machine Vision for real-world AI based Visual Inspection in manufacturing and Automobiles applications. My Research interests broadly include machine perception, 3D vision, and robotics, with a focus on advancing intelligent autonomous systems

Technical Skills

Skills: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Generative AI and Agentic AI

Frameworks & Tools: TensorFlow, PyTorch, Keras, Hugging Face, PaddleOCR

Edge AI: NVIDIA Jetson (Nano, Xavier), Model Optimization, ONNX, OpenVINO, TensorRt

Computer Vision and NLP: Object Detection & Tracking (ByteTrack), Image Segmentation, VisionTransformers, Synthetic Data Generation, YOLO Dinov2, Dinov3, CLIP, SAM, DETR, Qwen, LLaVA, Visual BERT, Gradcams, Stable Diffusion, GANs, OCR, Defect Detection, LLMs/VLMs/SLM

Image Processing: OpenCV, Pillow, Matplotlib, Image Thresholding, Image Registration, Image Enhancements and Scaling.

Agentic AI: Langchain, LangGraph, N8N, CrewAI, RAG,

EXPERIENCE

Orama Solutions

08/2023-Present

Senior Computer Vision Engineer

- Developed an AI-based Assembly Verification System for engine assembly of automobiles, focusing on missing part identification.
- Created AI-based Defect Detection Models using state-of-the-art Deep Learning algorithms and Image Processing to identify anomalies like scratches and dents in manufacturing industries, ensuring quality control.
- Implemented OCR Models to perform Optical Character Recognition on metal surfaces, enabling label identification and barcode matching.

Greentech ITs

06/2022 – 11/2022

Computer Vision Engineer

- Designed, developed, and implemented a complete state-of-the-art solution for Traffic Management Systems.
- Optimized and fine-tuned model architectures and training strategies to achieve superior performance in automated incidence detection on highways, contributing to increased road safety.
- Simplified transport solutions with automatic toll management systems.

GlobalLogic Technologies

05/2018- 01/2022

Senior Analyst

- Created Artificial Intelligence and Machine Learning-based applications to deliver valuable insights.
- Built algorithms based on Statistical Modeling, Machine Learning, and Deep Learning.

PROJECTS

Real-Time Automated Engine Assembly Block Inspection

- Developed and deployed a real-time machine vision system for automated inspection of vehicle engine blocks on production lines, ensuring 100% verification of critical component presence
- Achieved high precision and accuracy in detecting missing or misaligned parts, reducing manual inspection time and minimizing assembly defects in high-speed manufacturing environments.

Surface Defect Detection of Industrial Components Based on Vision

- Implemented a defect detection model to identify scratches and dents on industrial components.
- Applied various image processing techniques and heatmap visualization and Anomalib to enhance image quality and detect anomalies.

Mislabeled Packaging Inspection Using Optical Character Recognition

- Developed a system to detect the presence or absence of alphanumeric values on laser-engraved labels on cosmetic bottles.
- Utilized Paddle OCR to accurately recognize codes and characters, deploying the project on edge devices like Jetson Nano and Xavier.

Video Incident Detection System (VIDS)

- Developed a real-time traffic management system to detect and verify incidents using camera images.
- Enabled detection of stopped vehicles, speeding, fire, weather conditions, and traffic status.

EDUCATION

BTECH -ECE (Guru Jhambheshwar University of Science and Technology)

(2014-2017)